Zafaryab Haider

Ph.D. Candidate, Electrical & Computer Engineering, University of Maine | **Ph.D. expected March 2027**
Open to AI/ML research & engineering opportunities
Research Scientist | Applied Scientist | Research Engineer
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SUMMARY

AI security and trustworthy-ML researcher focused on adversarial and bit-flip attacks, RLHF training-signal integrity, and multi-agent LLM evaluation. Research **accepted at NeurIPS 2026** and **ACM CCS 2026** and **published at CVPR 2026**; 9 peer-reviewed papers published/accepted; **named inventor on 5 patent families**, including a joint UMaine/IN2FAB cross-PDK provisional.

SELECTED RESEARCH & PUBLICATIONS

NeurIPS 2026 · *Accepted* | **SMASH**: *Probing Speech Recognition Robustness via Semantically Targeted Bit Flips*
First author; led the project. Sparse, semantically targeted bit-flip attacks on Whisper and SeamlessM4T: a pooled median of **three INT8 bit flips** across 1,192 strict-success certificates; built the evaluation for reliability, transferability, collateral effects, and blinded human validation (human–LLM-judge agreement **kappa = 1.00**).

ACM CCS 2026 · *Accepted* | **(A)iSpy**: *Parasitic Trojans for Machine Learning Infrastructure*
Co-author. Parasitic-Trojan attacks on ML infrastructure; evaluates how security tools respond to them.

CVPR 2026 · *Published* | **DASH**: *Meta-Attack Framework for Effective & Stealthy Adversarial Examples*
Co-author. Composes existing attack primitives to jointly optimize adversarial effectiveness and stealth.

Scientific Reports (Nature Portfolio) 2025 · *Published* | *A Framework for Mitigating Malicious RLHF Feedback in LLM Training using Consensus Based Reward*
First author; led COBRA development. Raised reward accuracy to **about 73%** from 48.78% under malicious sentiment feedback; **about 23% over prior work at DF=10**; PCT filed.

IEEE CARS 2026 · *Accepted* | **ACT**: *Anchor-Calibrated Trust Estimation When Corrupted Sources Form Majority*
First author; led development and evaluation. Under corrupted-majority conditions, reversed reliability correlation from about −0.82 with agreement-only estimation to +0.82.

Under review | **BLADE**: *Bit-Flip Attacks for Semantic Misdirection in Quantized Image Captioning*
First author; led development. At a 20-bit budget, reached **48.80** penalized ASR vs. **20.95** Random+J, **10.25** PBS, and **8.20** GenBFA (BLIP2-OPT-6.7B/MS-COCO); U.S. provisional filed.

Under review | **HARP**: *Evaluating Agent Behavior After Compromises in Multi-agent Systems*
Second author. Benchmark for measuring how compromised agents amplify local failures into system-level harm in multi-agent LLMs; U.S. provisional filed.

RESEARCH ENGINEERING & EVALUATION

University of Maine | Graduate Research/Teaching Appointments | Orono, ME — August 2022 – Present

- **REWIRE** (Aug 2026 – Present): **leading the current research direction and experiments**; topology-aware world-model rollouts rank line-switching candidates before exact AC power-flow verification in Grid2Op.
- **Distributed LLM inference:** built layer-wise execution across heterogeneous Jetson/RTX hardware for models up to **15B parameters** under uneven compute.
- **Edge-AI testbed:** planned most architecture/deployment of a two-rack, **45-compute-device** testbed including **21 NVIDIA Jetsons**; led setup with another graduate researcher and guided an undergraduate.
- **Research interfaces:** built project-specific Streamlit/Python tools for human validation: a blinded annotation app for SMASH used by **13 annotators** and a separate evaluation tool for HARP used by **4 annotators**.
- **SentriAlign** (MIRTA 8.0, **$25,000**): **Co-Investigator & Technical Lead**; commercialization effort extending COBRA toward a training-signal integrity tool for LLM post-training.
- **AI/ML Consultant, ARCSIM** (Jun 2024 – May 2026, alongside graduate research): consulted university researchers on AI/ML methods; developed technical content and delivered seminars.

PRIOR EXPERIENCE

Aligarh Muslim University | Assistant Professor | Aligarh, India — January 2015 – August 2022
Designed and taught information security, networking, data structures, and C programming to 200+ students annually.

TECHNICAL SKILLS

ML & Security: PyTorch, Hugging Face, DeepSpeed, scikit-learn, NumPy, CuPy, Pandas; adversarial ML, bit-flip/fault injection, RLHF/reward modeling, LLM and agent evaluation
Systems & Research: CUDA, Docker, Linux, Slurm, Apptainer, multi-GPU/HPC, GCP, Jetson, Jupyter, Streamlit
Programming: Python, C/C++, SQL, Shell, Git, LaTeX

EDUCATION

Ph.D., Electrical and Computer Engineering, University of Maine, Orono, ME, USA — **Expected March 2027**
M.Tech., Computer Engineering, Aligarh Muslim University, India (GPA 9.33/10.0) — August 2011 – July 2013